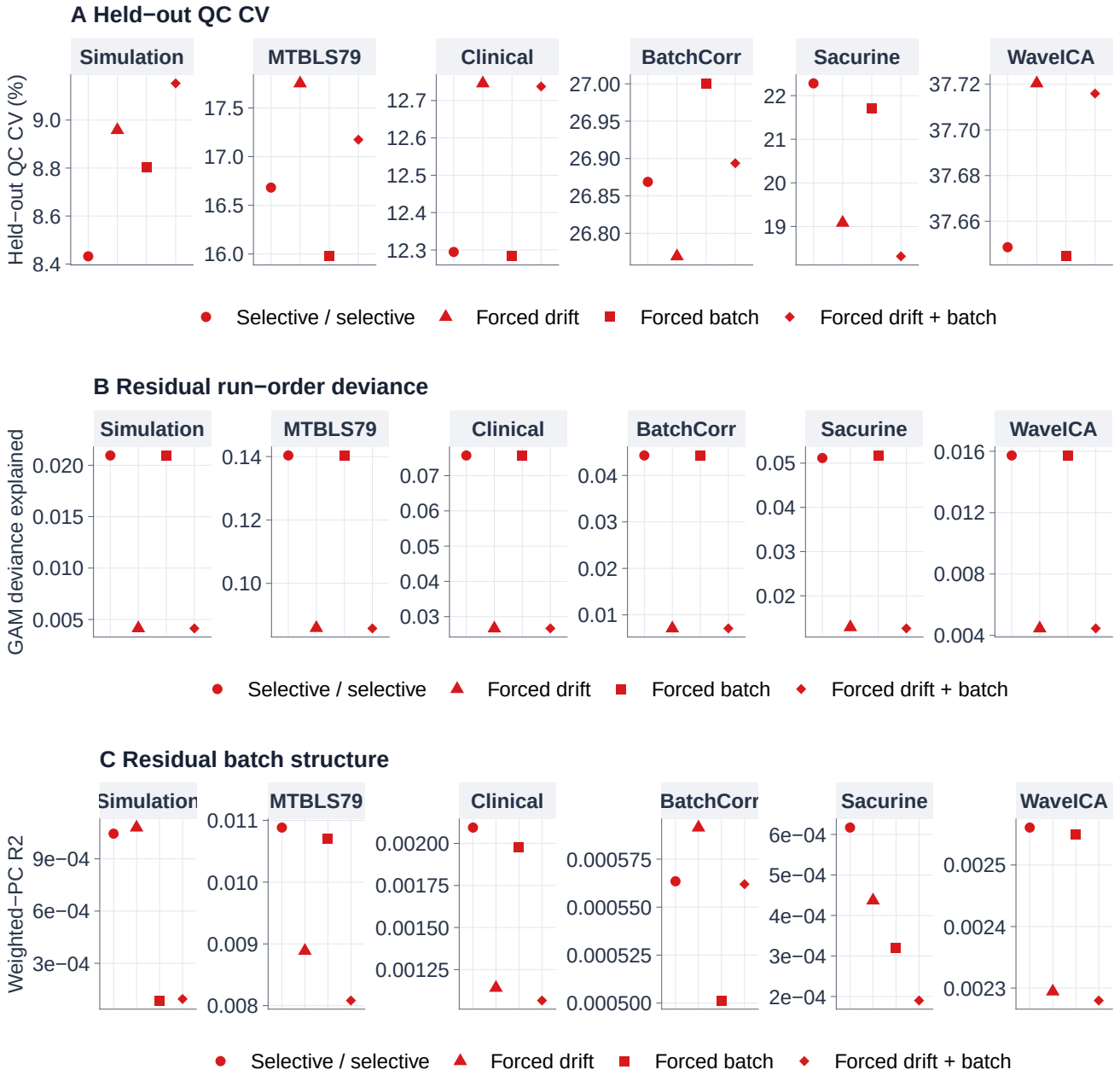
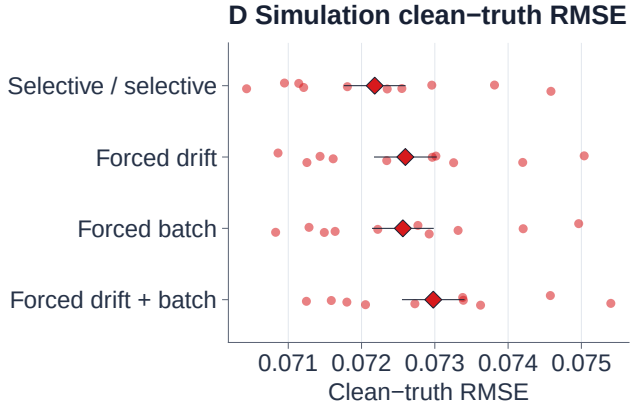
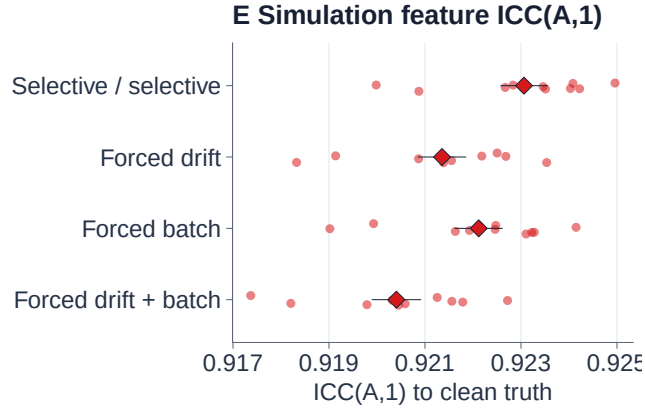
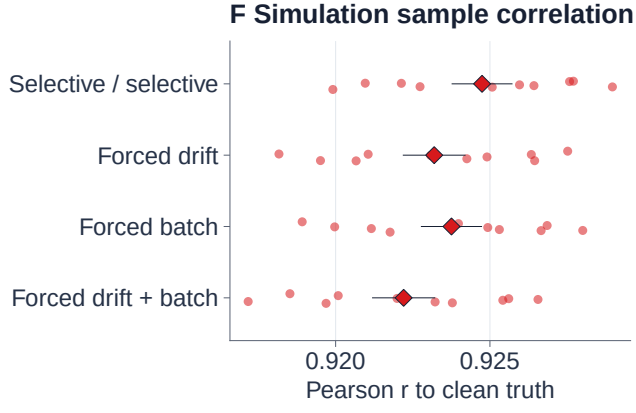

A Held−out QC CV
Held−out QC CV (%)
Simulation
MTBLS79
Clinical
BatchCorr
Sacurine
WaveICA
B Residual run−order deviance
GAM deviance explained
C Residual batch structure
Weighted−PC R2
Selective / selective
Forced drift
Forced batch
Forced drift + batch


D Simulation clean−truth RMSE
Selective / selective
Forced drift
Forced batch
Forced drift + batch
0.071
0.072
0.073
0.074
0.075
Clean−truth RMSE


E Simulation feature ICC(A,1)
Selective / selective
Forced drift
Forced batch
Forced drift + batch
0.917
0.919
0.921
0.923
0.925
ICC(A,1) to clean truth


F Simulation sample correlation
Selective / selective
Forced drift
Forced batch
Forced drift + batch
0.920
0.925
Pearson r to clean truth